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Here's your **Lab Monitoring System** report rewritten to be **concise (≈940 words)** while **preserving all core technical and academic content** — ideal for submission or presentation.

Lab Monitoring System

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Abstract

Efficient management of computer laboratories in educational institutions is essential yet challenging. Problems like unauthorized access, system overload, and inefficient resource usage reduce productivity and increase maintenance costs.
This project proposes a **Lab Monitoring System** that enables administrators to monitor system performance, user activity, and network usage in real time. The web-based platform provides a centralized dashboard displaying CPU, memory, and network usage, along with alerts and logs. Using modern web technologies and data handling methods, it ensures optimized lab performance and prevents misuse.

Keywords—Lab Monitoring, Resource Management, Real-time Tracking, Dashboard, Alerts, Network Monitoring

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I. INTRODUCTION

Computer laboratories play a vital role in education, offering students access to essential computing resources. However, manual monitoring methods are inefficient and fail to detect real-time issues such as system failures or resource abuse.
The **Lab Monitoring System** provides a centralized platform that continuously tracks multiple systems. It monitors CPU, memory, disk, and network parameters, displaying data through a real-time dashboard. The system allows administrators to detect issues early, analyze performance, and prevent misuse — improving both productivity and security.

II. PROBLEM STATEMENT

Educational institutions face issues like unmonitored usage, performance drops, and system downtime due to lack of visibility. Traditional manual monitoring is time-consuming and unreliable.
Existing solutions are often too technical or lack real-time features suitable for regular administrators.
The proposed system offers a **simple, automated, and centralized solution** to monitor lab systems, generate alerts, and visualize data efficiently.

III. SYSTEM DESCRIPTION

The Lab Monitoring System consists of the following key modules:

* **Admin Registration:** Administrators register and set monitoring preferences such as alert thresholds and notifications.
* **System Registration:** Each computer installs a lightweight client that sends system data to the server.
* **Real-time Monitoring:** The system tracks CPU, RAM, disk, and network usage in intervals and updates the dashboard live.
* **Automated Alerts:** Notifications are triggered when a system crosses set performance limits or goes offline.
* **Data Logging:** The system records performance data for future analysis and reports.
* **User Tracking:** User sessions and login durations are tracked to identify unauthorized access.
* **Remote Management:** Admins can send commands such as restart or shutdown to any connected PC.

IV. LITERATURE SURVEY

A. System Resource Monitoring:
Patel et al. (2019) proposed a monitoring tool to track CPU and memory data, enhancing system efficiency but adding network load.

B. User Activity Tracking:
Sharma & Gupta (2020) presented a framework to log sessions and software use, preventing misuse though requiring extra storage.

C. Centralized Dashboard:
Mehta et al. (2021) developed real-time dashboards that simplify monitoring but require setup time.

D. Automated Alerts:
Reddy & Kumar (2022) created an alert mechanism to reduce downtime but sometimes produced false positives.

E. Energy Optimization:
Das et al. (2020) designed power-efficient monitoring systems that lower energy use but depend on accurate idle detection.

V. METHODOLOGY

1. **Admin Login:** Admin defines alert limits and system settings.
2. **System Agent Setup:** Each system installs an agent that collects metrics.
3. **Data Collection:** Agents send data to the central server using REST APIs.
4. **Data Processing:** Server compares collected values with defined thresholds.
5. **Alerts and Notifications:** Emails or dashboard alerts are triggered automatically.
6. **Visualization:** Dashboard displays live charts and statistics using real-time updates.
7. **Reporting:** Usage history is compiled into reports for future optimization.

Fig. 1: System Flowchart

VI. IMPLEMENTATION

The implementation uses a **client-server model**. Each lab computer runs a background agent that periodically collects CPU, memory, and network data. This data is transmitted securely to a **Node.js/Express** server, which stores it in **MongoDB Atlas**.
The **frontend**, built using HTML, CSS, and Chart.js, fetches data via **REST APIs** and WebSockets to display real-time graphs.
Admins can log in to the dashboard to view system performance, receive alerts, and analyze usage trends.
The system thus ensures **real-time monitoring, alerting, and efficient management** without manual supervision.

VII. SYSTEM DIAGRAMS

• Architecture Diagram:
Each lab PC sends system data (CPU, memory, disk, and network) to the central server. The server processes and stores data in MongoDB and updates the dashboard using WebSockets.

• Use Case Diagram:
The client agent collects metrics, the backend handles data and alerts, and the frontend dashboard visualizes system health. Admins can view lab-wide or individual PC performance interactively.

VIII. CONCLUSION

The **Lab Monitoring System** provides an automated, reliable, and centralized way to manage computer laboratories.
It helps administrators monitor system health, track users, and optimize resources efficiently.

By shifting from manual to automated monitoring, it saves time, prevents misuse, and improves overall system performance.

Future Scope:

* Integration with mobile applications
* AI-based predictive maintenance
* Expansion to campus-wide infrastructure monitoring

IX. REFERENCES

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